

Δn

Calcul de Δn

$$\Delta n = \sqrt{\left(\frac{\partial n}{\partial c}\right)^2 \Delta c^2}$$

$$\Delta n = \left| \frac{\partial n}{\partial c} \right| \Delta c$$

$$\Delta c = \left| \frac{\partial c}{\partial i} \Delta i \right| + \left| \frac{\partial c}{\partial D} \Delta D \right|$$

Estimons $\Delta i \approx 0,1 \text{ cm}$

et $\Delta D = 0,1 \text{ cm}$

$$\frac{\partial c}{\partial i} = -\frac{\lambda D}{i^2} = -0,0098$$

$$\frac{\partial c}{\partial D} = \frac{1}{i} = 7,91 \cdot 10^{-5}$$

$$\Delta c = 9,8 \cdot 10^{-6} + 7,91 \cdot 10^{-5} \times 10^{-3}$$

$$\Delta c \approx 9,9 \cdot 10^{-6}$$

 Δn

$$\frac{\partial n}{\partial c} = -\frac{1}{c^2} = -\frac{1}{(7,91 \cdot 10^5)^2} = 15825853,7$$

$$\Delta n = \left| \frac{\partial n}{\partial c} \Delta c \right| = 1582 \text{ traits par mètre}$$

$$\Delta n = 1,582 \approx 2 \text{ traits par mm}$$

$$n \pm \Delta n = 13 \pm 2 \text{ traits par mm}$$